

Practical 2

In [272... `import pandas as pd`

In [273... `import matplotlib.pyplot as mt`

In [274... `import seaborn as sns`

In [275... `import numpy as np`

In [276... `import matplotlib.pyplot as plt`

In [277... `df=pd.read_csv("academic_performance.csv")`

In [278... `#incosistencies`
`for col in ['Maths','Science','English','Attendance']:`
 `print("Incosistent data in",col)`
 `print(df[(df[col]<0) | (df[col]>100)])`

Incosistent data in Maths

	Student_ID	Maths	Science	English	Attendance
6	7	150.0	85.0	65.0	88.0
20	21	110.0	85.0	88.0	90.0

Incosistent data in Science

	Student_ID	Maths	Science	English	Attendance
7	8	72.0	400.0	78.0	92.0

Incosistent data in English

Empty DataFrame

Columns: [Student_ID, Maths, Science, English, Attendance]

Index: []

Incosistent data in Attendance

	Student_ID	Maths	Science	English	Attendance
17	18	92.0	95.0	98.0	105.0

In [279... `for col in ['Maths','Science','English','Attendance']:`
 `df.loc[(df[col]<0) | (df[col]>100),col]=np.nan`

In [280... `df`

Out[280...

	Student_ID	Maths	Science	English	Attendance
0	1	78.0	82.0	75.0	85.0
1	2	85.0	79.0	80.0	90.0
2	3	NaN	88.0	70.0	78.0
3	4	95.0	NaN	85.0	95.0
4	5	40.0	35.0	NaN	60.0
5	6	88.0	90.0	88.0	NaN
6	7	NaN	85.0	65.0	88.0
7	8	72.0	NaN	78.0	92.0
8	9	66.0	70.0	68.0	75.0
9	10	55.0	60.0	58.0	65.0
10	11	90.0	92.0	94.0	98.0
11	12	48.0	52.0	50.0	55.0
12	13	82.0	78.0	80.0	82.0
13	14	76.0	NaN	74.0	80.0
14	15	NaN	65.0	60.0	70.0
15	16	68.0	72.0	70.0	77.0
16	17	85.0	88.0	90.0	89.0
17	18	92.0	95.0	98.0	NaN
18	19	35.0	30.0	40.0	45.0
19	20	60.0	58.0	62.0	68.0
20	21	NaN	85.0	88.0	90.0
21	22	73.0	75.0	72.0	NaN
22	23	80.0	82.0	78.0	85.0
23	24	65.0	67.0	70.0	72.0
24	25	90.0	88.0	NaN	95.0
25	26	55.0	50.0	48.0	60.0
26	27	78.0	76.0	80.0	83.0
27	28	82.0	85.0	88.0	87.0
28	29	95.0	98.0	96.0	99.0
29	30	20.0	25.0	30.0	35.0

```
In [281... #column-wise inconsitent values  
df.isnull().sum()
```

```
Out[281... Student_ID    0  
Maths         4  
Science       3  
English       2  
Attendance    3  
dtype: int64
```

```
In [282... #replace null values with 0  
#df.fillna(0)
```

```
In [283... #replacing Nan with NULL  
#df.replace(np.nan, "NULL")
```

```
In [284... df["Maths"].mean()
```

```
Out[284... np.float64(71.26923076923077)
```

```
In [285... #replacing NaN with mean of column  
df["Maths"] = df["Maths"].fillna(df["Maths"].mean()).round(2)
```

```
In [286... df
```

Out[286...

	Student_ID	Maths	Science	English	Attendance
0	1	78.00	82.0	75.0	85.0
1	2	85.00	79.0	80.0	90.0
2	3	71.27	88.0	70.0	78.0
3	4	95.00	NaN	85.0	95.0
4	5	40.00	35.0	NaN	60.0
5	6	88.00	90.0	88.0	NaN
6	7	71.27	85.0	65.0	88.0
7	8	72.00	NaN	78.0	92.0
8	9	66.00	70.0	68.0	75.0
9	10	55.00	60.0	58.0	65.0
10	11	90.00	92.0	94.0	98.0
11	12	48.00	52.0	50.0	55.0
12	13	82.00	78.0	80.0	82.0
13	14	76.00	NaN	74.0	80.0
14	15	71.27	65.0	60.0	70.0
15	16	68.00	72.0	70.0	77.0
16	17	85.00	88.0	90.0	89.0
17	18	92.00	95.0	98.0	NaN
18	19	35.00	30.0	40.0	45.0
19	20	60.00	58.0	62.0	68.0
20	21	71.27	85.0	88.0	90.0
21	22	73.00	75.0	72.0	NaN
22	23	80.00	82.0	78.0	85.0
23	24	65.00	67.0	70.0	72.0
24	25	90.00	88.0	NaN	95.0
25	26	55.00	50.0	48.0	60.0
26	27	78.00	76.0	80.0	83.0
27	28	82.00	85.0	88.0	87.0
28	29	95.00	98.0	96.0	99.0
29	30	20.00	25.0	30.0	35.0

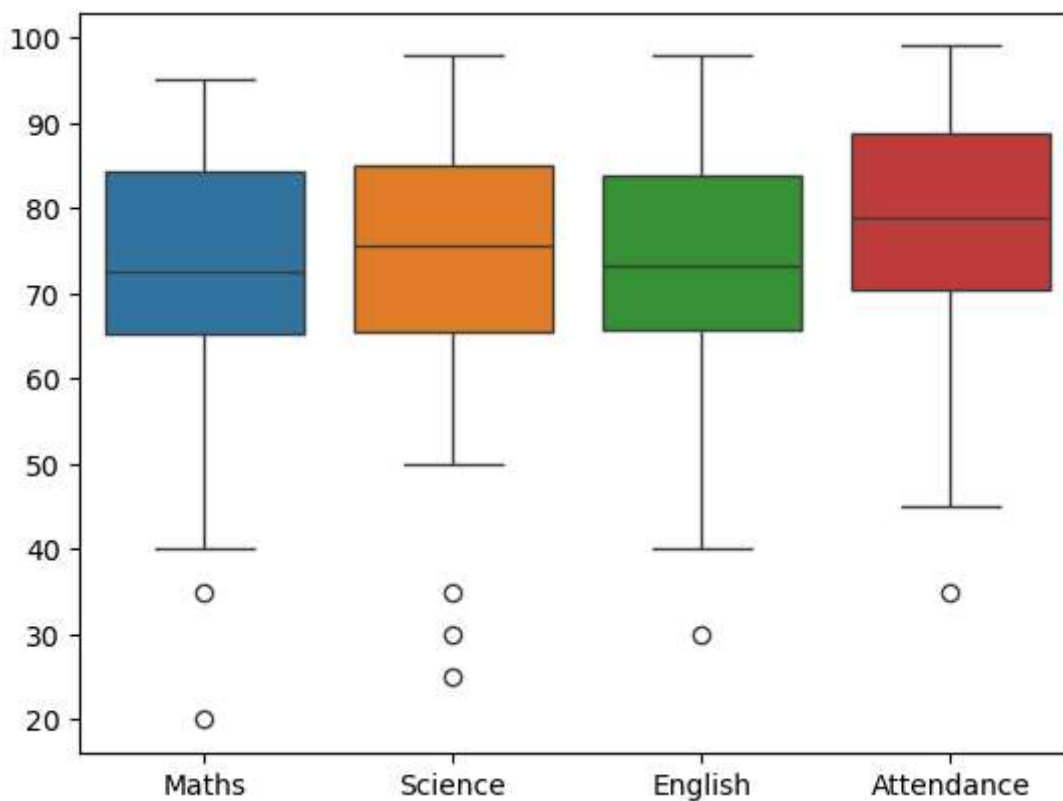
```
In [287... #replacing NaN with mean of column  
for col in ['Maths', 'Science', 'English', 'Attendance']:  
    df[col]=df[col].fillna(df[col].mean()).round(2)
```

```
In [288... df
```

Out[288...

	Student_ID	Maths	Science	English	Attendance
0	1	78.00	82.00	75.00	85.0
1	2	85.00	79.00	80.00	90.0
2	3	71.27	88.00	70.00	78.0
3	4	95.00	72.22	85.00	95.0
4	5	40.00	35.00	72.68	60.0
5	6	88.00	90.00	88.00	77.7
6	7	71.27	85.00	65.00	88.0
7	8	72.00	72.22	78.00	92.0
8	9	66.00	70.00	68.00	75.0
9	10	55.00	60.00	58.00	65.0
10	11	90.00	92.00	94.00	98.0
11	12	48.00	52.00	50.00	55.0
12	13	82.00	78.00	80.00	82.0
13	14	76.00	72.22	74.00	80.0
14	15	71.27	65.00	60.00	70.0
15	16	68.00	72.00	70.00	77.0
16	17	85.00	88.00	90.00	89.0
17	18	92.00	95.00	98.00	77.7
18	19	35.00	30.00	40.00	45.0
19	20	60.00	58.00	62.00	68.0
20	21	71.27	85.00	88.00	90.0
21	22	73.00	75.00	72.00	77.7
22	23	80.00	82.00	78.00	85.0
23	24	65.00	67.00	70.00	72.0
24	25	90.00	88.00	72.68	95.0
25	26	55.00	50.00	48.00	60.0
26	27	78.00	76.00	80.00	83.0
27	28	82.00	85.00	88.00	87.0
28	29	95.00	98.00	96.00	99.0
29	30	20.00	25.00	30.00	35.0

```
In [289... sns.boxplot(data=df[['Maths','Science','English','Attendance']])
plt.show()
```



```
In [290... Q1=df['Maths'].quantile(0.25)
```

```
In [291... Q1
```

```
Out[291... np.float64(65.25)
```

```
In [292... Q2=df['Maths'].quantile(0.50)
```

```
In [293... Q2
```

```
Out[293... np.float64(72.5)
```

```
In [294... Q3=df['Maths'].quantile(0.75)
```

```
In [295... Q3
```

```
Out[295... np.float64(84.25)
```

```
In [296... IQR=Q3-Q1
lower=Q1-1.5*IQR
print("IQR",IQR)
print("LOW",lower)
upper=Q3+1.5*IQR
print("Upper",upper)
```

IQR 19.0
LOW 36.75
Upper 112.75

```
In [297... cols=['Maths','Science','English','Attendance']
```

```
In [298... all_Q1=df[cols].quantile(0.25)
```

```
In [299... all_Q1
```

```
Out[299... Maths      65.25  
Science      65.50  
English      65.75  
Attendance   70.50  
Name: 0.25, dtype: float64
```

```
In [300... all_Q3=df[cols].quantile(0.75)
```

```
In [301... all_Q3
```

```
Out[301... Maths      84.25  
Science      85.00  
English      83.75  
Attendance   88.75  
Name: 0.75, dtype: float64
```

```
In [302... all_IQR=all_Q3-all_Q1
```

```
In [303... all_IQR
```

```
Out[303... Maths      19.00  
Science      19.50  
English      18.00  
Attendance   18.25  
dtype: float64
```

```
In [304... lower_bound=all_Q1-1.5*all_IQR
```

```
In [305... lower_bound
```

```
Out[305... Maths      36.750  
Science      36.250  
English      38.750  
Attendance   43.125  
dtype: float64
```

```
In [306... upper_bound=all_Q3+1.5*all_IQR
```

```
In [307... upper_bound
```



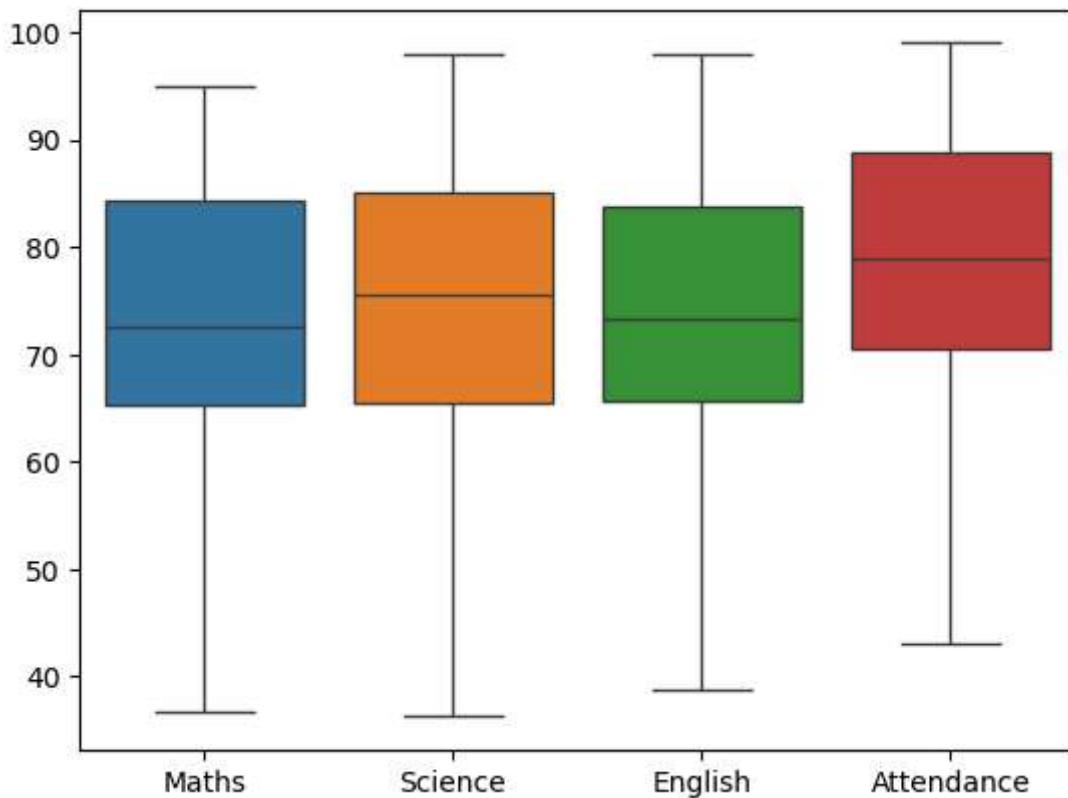
```
Out[307... Maths      112.750
Science    114.250
English    110.750
Attendance 116.125
dtype: float64
```

```
In [308... #replace with median
#for col in ['Maths','Science','English','Attendance']:
# df.loc[(df[col]<lower_bound[col]) | (df[col]>upper_bound[col]),col]=df[col].me
```

```
In [309... #sns.boxplot(data=df[['Maths','Science','English','Attendance']])
#plt.show()
```

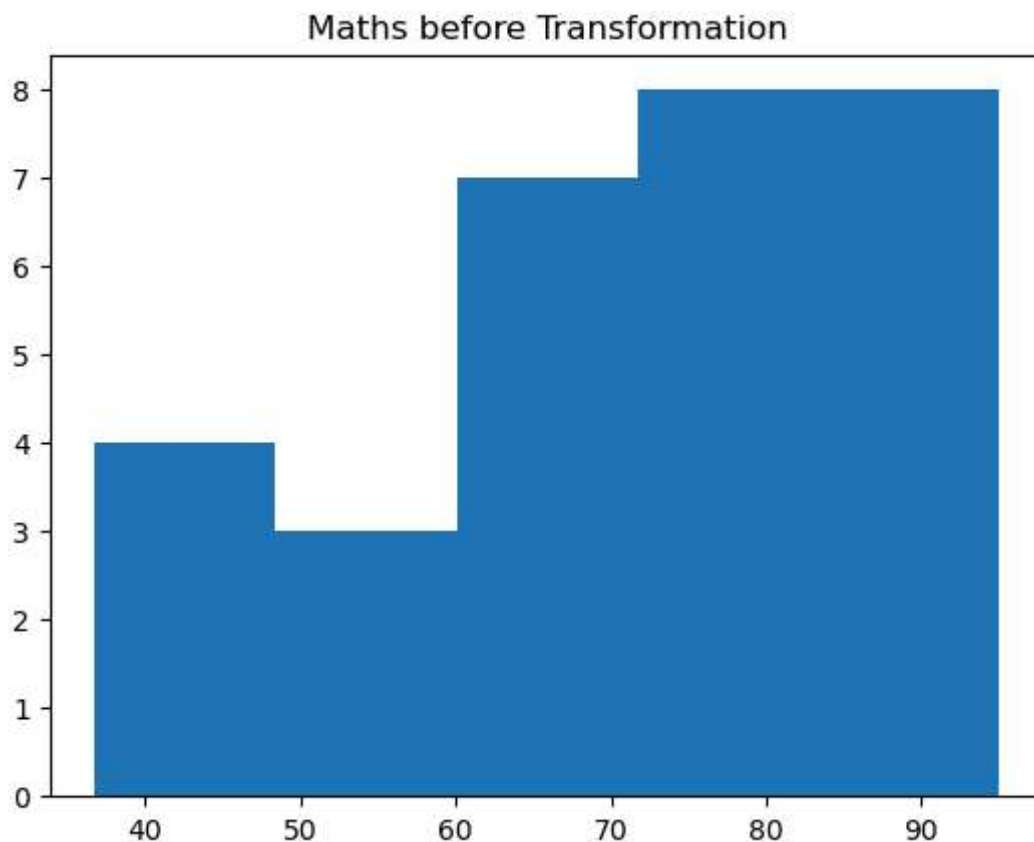
```
In [310... for col in ['Maths','Science','English','Attendance']:
df.loc[(df[col]<lower_bound[col]),col]=lower_bound[col]
df.loc[(df[col]>upper_bound[col]),col]=upper_bound[col]
```

```
In [311... sns.boxplot(data=df[['Maths','Science','English','Attendance']])
plt.show()
```



Transformation

```
In [312... #skewness
mt.hist(df['Maths'],bins=5)
mt.title("Maths before Transformation")
mt.show()
```



```
In [319... #negative skewness  
df['Maths'].skew()
```

```
Out[319... np.float64(-0.7157883780071774)
```

```
In [320... #df['Maths Transformed']=np.Log(df['Maths'])
```

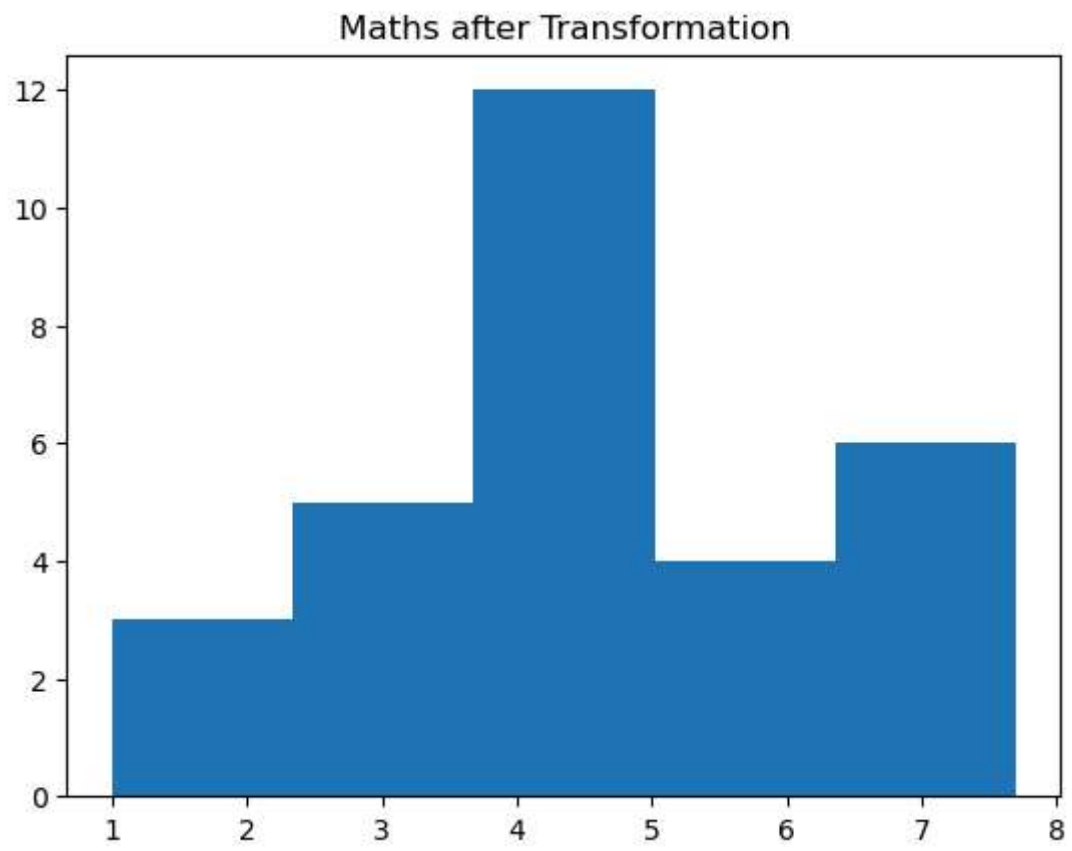
```
In [321... #df['Maths Transformed'].skew()
```

```
In [322... max=df['Maths'].max()  
df['Maths reflected']=max+1-df['Maths']  
df['Maths Transformed']=np.sqrt(df['Maths reflected'])
```

```
In [323... df['Maths Transformed'].skew()
```

```
Out[323... np.float64(-0.14533851929568667)
```

```
In [324... mt.hist(df['Maths Transformed'],bins=5)  
mt.title("Maths after Transformation")  
mt.show()
```



In []: